Implementation of Adversarial Audio Attacks using ALIF/TASER and Whisper ASR

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1 Abstract

Automatic Speech Recognition (ASR) systems are widely used in smart assistants, banking systems, and voice-controlled applications. However, these systems are vulnerable to adversarial audio attacks.

This project implements a simplified version of the research paper *"ALIF: Low-Cost Adversarial Audio Attacks on Black-Box Speech Platforms Using Linguistic Features"* using the TASER repository and Whisper ASR.

The system generates adversarial audio by introducing small perturbations into speech signals. These perturbations are nearly imperceptible to humans but cause speech recognition systems to generate modified transcriptions.

The project demonstrates adversarial audio generation, transcription comparison, Word Error Rate (WER) evaluation, waveform visualization, and spectrogram analysis.

2 Introduction

Automatic Speech Recognition (ASR) converts spoken language into text using machine learning and deep neural networks.

Modern ASR systems such as:

- Alexa
- Siri
- Google Assistant
- Whisper ASR

are widely used in real-world applications.

Despite their effectiveness, ASR systems are vulnerable to adversarial attacks where attackers manipulate audio inputs to produce incorrect transcription outputs.

The ALIF/TASER framework demonstrates how small perturbations added to speech signals can fool ASR systems in a black-box setting.

3 Objectives

The objectives of this project are:

- Study adversarial audio attacks on ASR systems
- Implement a simplified ALIF/TASER framework
- Integrate Whisper ASR
- Generate adversarial audio samples
- Compare original and adversarial transcripts
- Evaluate attack success using Word Error Rate (WER)
- Generate waveform and spectrogram graphs

4 System Architecture

Input Audio
$\downarrow$
Whisper ASR
$\downarrow$
Original Transcript
$\downarrow$
Adversarial Perturbation
$\downarrow$
Adversarial Audio
$\downarrow$
Whisper ASR
$\downarrow$
Modified Transcript

5 Methodology

5.1 Input Audio

A speech audio file is used as input for the ASR system.

5.2 Speech Recognition

The Whisper ASR model converts the speech signal into text transcription.

5.3 Adversarial Perturbation

Small perturbation noise is added to the audio signal.

Mathematically:

$$x_{adv} = x + \delta$$

Where:

- x = Original audio
- δ = Perturbation noise
- x_{adv} = Adversarial audio

5.4 Adversarial Audio Generation

The modified audio becomes adversarial audio which sounds nearly identical to humans but manipulates ASR outputs.

5.5 Transcription Comparison

The adversarial audio is again processed by Whisper ASR and compared with the original transcript.

6 Tools and Technologies Used

Component	Technology
Programming Language	Python
Speech Recognition	Whisper ASR
Framework	PyTorch
Audio Processing	Librosa
Visualization	Matplotlib
Evaluation Metric	JiWER
Platform	Linux / WSL

7 Experimental Results

7.1 Original Transcript

Scientists at the Sur laboratory say they have discovered a new particle.

7.2 Adversarial Transcript

Scientists at the Surin Laboratory say they have discovered a new particle.

7.3 Word Error Rate

$$WER = 0.1667$$

The result demonstrates successful adversarial manipulation of the ASR system.

8 Result Graphs

8.1 Waveform Graph

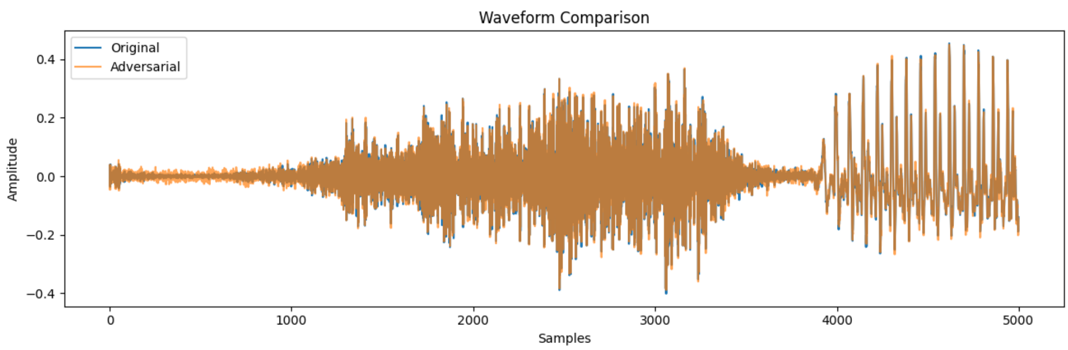

Figure 1: Waveform comparison of adversarial audio

8.2 Adversarial Spectrogram

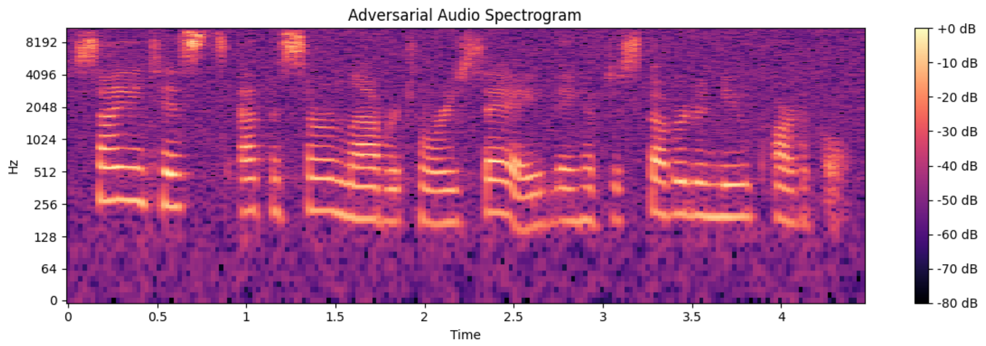

Figure 2: Spectrogram visualization of adversarial audio

9 Generated Outputs

The following files were generated during implementation:

File	**Description**
adversarial.wav	Adversarial audio sample
waveform.png	Waveform comparison graph
spectrogram.png	Adversarial spectrogram

10 Novelty Work Done

The original TASER repository required external ASR APIs and complex configurations. In this project, the following improvements were implemented:

- Whisper ASR was integrated locally
- Simplified adversarial perturbation generation was implemented
- Automatic waveform visualization was added
- Spectrogram generation was implemented
- Word Error Rate (WER) evaluation was included
- The implementation was optimized for CPU-based execution

11 Advantages

- Demonstrates vulnerabilities in ASR systems
- Black-box adversarial attack implementation
- Lightweight and reproducible
- Useful for AI security research

12 Limitations

- Simplified perturbation method
- Limited dataset usage
- CPU-based execution increases runtime

13 Conclusion

This project successfully implemented a simplified adversarial audio attack framework based on the ALIF/TASER research paper using Whisper ASR.

Experimental results demonstrated that small perturbations added to speech signals can manipulate speech recognition outputs while remaining nearly imperceptible to humans.

The project highlights security vulnerabilities in modern ASR systems and emphasizes the importance of developing robust speech recognition defenses.

14 References

1. ALIF: Low-Cost Adversarial Audio Attacks on Black-Box Speech Platforms Using Linguistic Features.
2. TASER GitHub Repository.
3. OpenAI Whisper Documentation.
4. PyTorch Documentation.
5. Librosa Audio Processing Library Documentation.